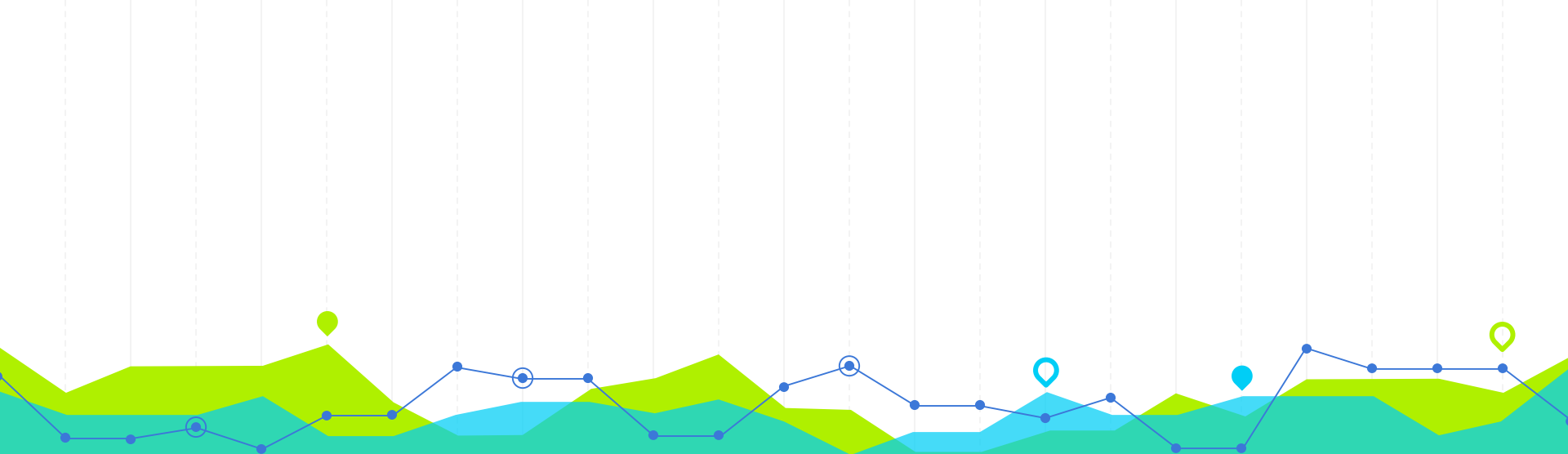




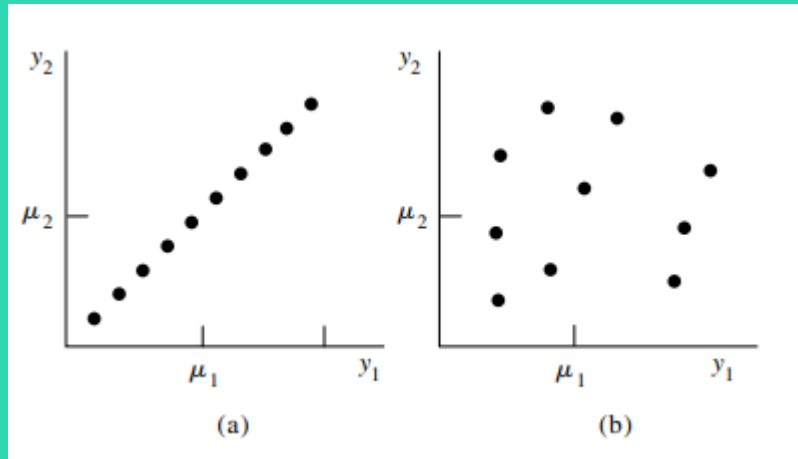
Multivariate Probability Distributions

Covariance and Conditional Expectation of function of
random variables

Covariance



Think of the dependence of two random variables Y_1 and Y_2 as implying that one variable say, Y_1 either increases or decreases as Y_2 changes.
We will confine our attention to two measures of dependence:
the covariance between two random variables
their correlation coefficient



Dependent and independent observations for (y_1, y_2)

Suppose that we knew the values of $E(Y_1) = \mu_1$ and $E(Y_2) = \mu_2$ and located this point on the graph in previous Figure.

Now locate a plotted point, (y_1, y_2) , on Figure (a) and measure the deviations $(y_1 - \mu_1)$ and $(y_2 - \mu_2)$.

Both deviations assume the same algebraic sign for any point, (y_1, y_2) , and their product $(y_1 - \mu_1)(y_2 - \mu_2)$ is positive.



Points to the right of μ_1 yield pairs of positive deviations; points to the left produce pairs of negative deviations; and the average of the product of the deviations $(y_1 - \mu_1)(y_2 - \mu_2)$ is large and positive.

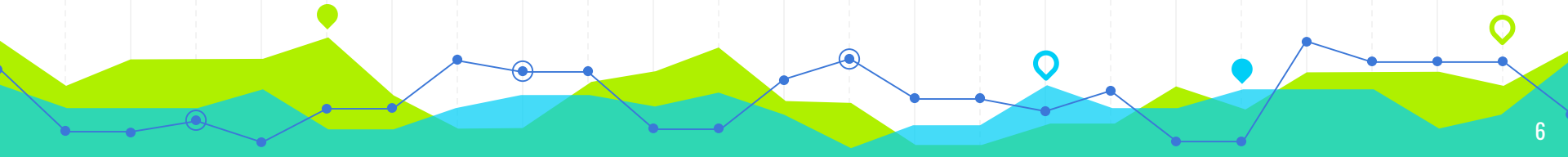
If the linear relation indicated in Figure (a) had sloped **downward to the right**, all corresponding pairs of deviations would have been of the opposite sign, and the average value of $(y_1 - \mu_1)(y_2 - \mu_2)$ would have been a large negative number.



The situation just described does not occur for Figure (b), where little dependence exists between Y_1 and Y_2 .

Their corresponding deviations $(y_1 - \mu_1)$ and $(y_2 - \mu_2)$ will assume the same algebraic sign for some points and opposite signs for others.

Thus, the product $(y_1 - \mu_1)(y_2 - \mu_2)$ will be positive for some points, negative for others, and will average to some value near zero.



The average value of $(Y_1 - \mu_1)(Y_2 - \mu_2)$ provides a measure of the linear dependence between Y_1 and Y_2 .

This quantity, $E[(Y_1 - \mu_1)(Y_2 - \mu_2)]$, is called the **covariance** of Y_1 and Y_2 .



DEFINITION 10

If Y_1 and Y_2 are random variables with means μ_1 and μ_2 , respectively, the covariance of Y_1 and Y_2 is

$$\text{Cov}(Y_1, Y_2) = E[(Y_1 - \mu_1)(Y_2 - \mu_2)] .$$

PROOF

From Theorem 8 the expected value of a sum is equal to the sum of the expected values; and from Theorem 7, the expected value of a constant times a function of random variables is the constant times the expected value.

Thus, $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - \mu_1 E(Y_2) - \mu_2 E(Y_1) + \mu_1 \mu_2$.

Because $E(Y_1) = \mu_1$ and $E(Y_2) = \mu_2$,
it follows that

$$\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = E(Y_1 Y_2) - \mu_1 \mu_2.$$

- The **larger the absolute value** of the covariance of Y_1 and Y_2 , the **greater the linear dependence** between Y_1 and Y_2 .
- **Positive values** indicate that Y_1 increases as Y_2 increases; **negative values** indicate that Y_1 decreases as Y_2 increases.
- A **zero value** of the covariance indicates that the variables are **uncorrelated** and that there is **no linear dependence between Y_1 and Y_2** .



It is **difficult to employ** the **covariance** as an **absolute measure** of dependence because **its value depends upon the scale of measurement**.

As a result, it is difficult to determine at first glance whether a particular covariance is large or small.

This **problem can be eliminated by** standardizing its value and **using the correlation coefficient, ρ** , a quantity related to the covariance and defined as

$$\rho = \frac{\text{Cov}(Y_1, Y_2)}{\sigma_1 \sigma_2}$$

where σ_1 and σ_2 are the standard deviations of Y_1 and Y_2 , respectively.

The sign of the correlation coefficient is the same as the sign of the covariance.

$\rho > 0 \Rightarrow Y_2$ increases as Y_1 increases

$\rho = +1 \Rightarrow$ perfect correlation, with all points falling on a straight line with positive slope.

$\rho = 0 \Rightarrow$ zero covariance and no correlation.

$\rho < 0 \Rightarrow$ a decrease in Y_2 as Y_1 increases

$\rho = -1 \Rightarrow$ perfect correlation, with all points falling on a straight line with negative slope.



EXAMPLE 19

Y_1 and Y_2 have joint density function given by

Find the covariance between Y_1 and Y_2 .

$$f(y_1, y_2) = \begin{cases} 3y_1, & 0 \leq y_2 \leq y_1 \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

EXAMPLE 20

Let Y_1 and Y_2 have joint density given by

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the covariance between the amount in stock Y_1 and amount of sales Y_2 .

THEOREM 11

If Y_1 and Y_2 are independent random variables, then

$$\text{Cov}(Y_1, Y_2) = 0.$$

Thus, independent random variables must be uncorrelated.

PROOF

Theorem 10 establishes that

$$\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - \mu_1 \mu_2 .$$

Because Y_1 and Y_2 are independent, Theorem 9 implies that

$$E(Y_1 Y_2) = E(Y_1)E(Y_2) = \mu_1 \mu_2 ,$$

and the desired result follows immediately.

EXAMPLE 21

Let Y_1 and Y_2 be discrete random variables with joint probability distribution as shown in following table. Show that Y_1 and Y_2 are dependent but have zero covariance.

y_2	y_1		
	-1	0	+1
-1	1/16	3/16	1/16
0	3/16	0	3/16
+1	1/16	3/16	1/16

Conditional Expectations



DEFINITION 11

If Y_1 and Y_2 are any two random variables, the conditional expectation of $g(Y_1)$, given that $Y_2 = y_2$, is defined to be

$$E(g(Y_1) | Y_2 = y_2) = \int_{-\infty}^{\infty} g(y_1) f(y_1 | y_2) dy_1$$

if Y_1 and Y_2 are jointly continuous and

$$E(g(Y_1) | Y_2 = y_2) = \sum_{\text{all } y_1} g(y_1) p(y_1 | y_2)$$

if Y_1 and Y_2 are jointly discrete.

EXAMPLE 22

Random variables Y_1 and Y_2 , where the joint density function is given by

$$f(y_1, y_2) = \begin{cases} 1/2, & 0 \leq y_1 \leq y_2 \leq 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the conditional expectation of the amount of sales, Y_1 , given that $Y_2 = 1.5$.

THEOREM 12

Let Y_1 and Y_2 denote random variables. Then

$$E(Y_1) = E[E(Y_1 | Y_2)],$$

where on the right-hand side the inside expectation is with respect to the conditional distribution of Y_1 given Y_2 and the outside expectation is with respect to the distribution of Y_2 .

PROOF

Suppose that Y_1 and Y_2 are jointly continuous with joint density function $f(y_1, y_2)$ and marginal densities $f_1(y_1)$ and $f_2(y_2)$, respectively. Then

$$\begin{aligned} E(Y_1) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y_1 f(y_1, y_2) dy_1 dy_2 \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y_1 f(y_1 | y_2) f_2(y_2) dy_1 dy_2 \\ &= \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} y_1 f(y_1 | y_2) dy_1 \right\} f_2(y_2) dy_2 \\ &= \int_{-\infty}^{\infty} E(Y_1 | Y_2 = y_2) f_2(y_2) dy_2 = E[E(Y_1 | Y_2)]. \end{aligned}$$

The proof is similar for the discrete case.

EXAMPLE 23

A quality control plan for an assembly line involves sampling $n = 10$ finished items per day and counting Y , the number of defectives. If p denotes the probability of observing a defective, then Y has a binomial distribution, assuming that a large number of items are produced by the line. But p varies from day to day and is assumed to have a uniform distribution on the interval from 0 to $1/4$. Find the expected value of Y .

Conditional Variance

The conditional variance of Y_1 given $Y_2 = y_2$ is defined by analogy with an ordinary variance, again using the conditional density or probability function of Y_1 given $Y_2 = y_2$ in place of the ordinary density or probability function of Y_1 .

That is,

$$V(Y_1 | Y_2 = y_2) = E(Y_1^2 | Y_2 = y_2) - [E(Y_1 | Y_2 = y_2)]^2.$$

THEOREM 13

Let Y_1 and Y_2 denote random variables. Then

$$V(Y_1) = E [V(Y_1 | Y_2)] + V [E(Y_1 | Y_2)].$$

PROOF

As previously indicated, $V(Y_1 | Y_2)$ is given by

$$V(Y_1 | Y_2) = E(Y_1^2 | Y_2) - [E(Y_1 | Y_2)]^2$$

and

$$E[V(Y_1 | Y_2)] = E[E(Y_1^2 | Y_2)] - E\{[E(Y_1 | Y_2)]^2\}.$$

By definition,

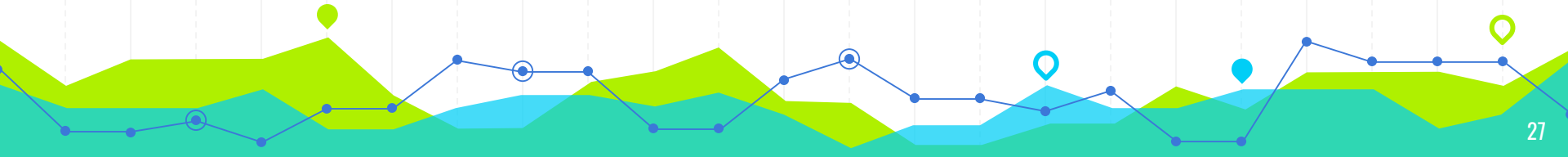
$$V[E(Y_1 | Y_2)] = E\{[E(Y_1 | Y_2)]^2\} - \{E[E(Y_1 | Y_2)]\}^2.$$

The variance of Y_1 is

$$\begin{aligned} V(Y_1) &= E[Y_1^2] - [E(Y_1)]^2 \\ &= E\{E[Y_1^2 | Y_2]\} - \{E[E(Y_1 | Y_2)]\}^2 \\ &= E\{E[Y_1^2 | Y_2]\} - E\{[E(Y_1 | Y_2)]^2\} + E\{[E(Y_1 | Y_2)]^2\} \\ &\quad - \{E[E(Y_1 | Y_2)]\}^2 \\ &= E[V(Y_1 | Y_2)] + V[E(Y_1 | Y_2)]. \end{aligned}$$

EXAMPLE 24

Find the variance of Y .



Q1

The random variables Y_1 and Y_2 are such that $E(Y_1) = 4$, $E(Y_2) = -1$, $V(Y_1) = 2$ and $V(Y_2) = 8$.

a What is $\text{Cov}(Y_1, Y_1)$?

b Assuming that the means and variances are correct, as given, is it possible that $\text{Cov}(Y_1, Y_2) = 7$? [Hint: If $\text{Cov}(Y_1, Y_2) = 7$, what is the value of ρ , the coefficient of correlation?]

c Assuming that the means and variances are correct, what is the largest possible value for $\text{Cov}(Y_1, Y_2)$? If $\text{Cov}(Y_1, Y_2)$ achieves this largest value, what does that imply about the relationship between Y_1 and Y_2 ?

d Assuming that the means and variances are correct, what is the smallest possible value for $\text{Cov}(Y_1, Y_2)$? If $\text{Cov}(Y_1, Y_2)$ achieves this smallest value, what does that imply about the relationship between Y_1 and Y_2 ?

Q2

Assume that Y denotes the number of bacteria per cubic centimeter in a particular liquid and that Y has a Poisson distribution with parameter λ . Further assume that λ varies from location to location and has a gamma distribution with parameters α and β , where α is a positive integer. If we randomly select a location, what is the

- a** expected number of bacteria per cubic centimeter?
- b** standard deviation of the number of bacteria per cubic centimeter?